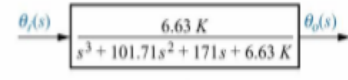
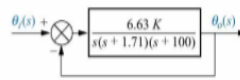
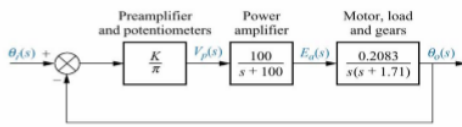
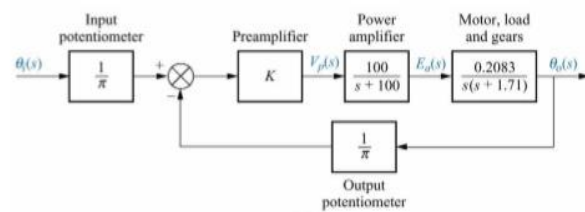


DESIGN WITH ROOT LOCUS

Consider the example of an antenna position control system as shown. The required position is set on the input potentiometer and the actual position is sensed by the output potentiometer. The block diagram is modified as shown



With the preamplifier gain $K = 64.21$, let us find out the **closed-loop poles** and the system's dynamic specifications.

$$p_i = -100, -0.83 \pm j1.89$$

Since the third pole $s = -100$ is far to the left of origin, the **dominant poles** are $-0.83 \pm j1.89$

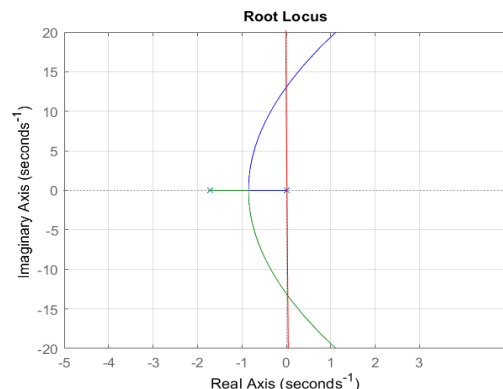
$$-\zeta\omega_n = -0.83, \quad \omega_d = 1.89 \frac{\text{rad}}{\text{s}} \gg \gg \gg \quad \zeta = 0.402, \quad \omega_n = 2.06 \frac{\text{rad}}{\text{s}}, \quad \theta = \cos^{-1} \zeta = 66.3^\circ$$

$$t_r = \frac{\pi - \theta}{\omega_d} = 1.08 \text{ s}, \quad \%OS = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 25\%, \quad t_p = \frac{\pi}{\omega_d} = 1.6 \text{ s}, \quad 2\% t_s = \frac{4}{\zeta\omega_n} = 4.8 \text{ s}$$

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = 2.5$$

$$KG(s)H(s) = \frac{6.63K}{s(s+1.71)(s+100)}$$

The poles of $G(s)H(s)$ are $s = 0, -1.71, -100$. Construct the root loci. The portion of the root loci close to the origin is shown. The third segment of the loci will start from -100 and extend to $-\infty$ which is not shown in the figure. If we draw a line at an angle $\theta = \cos^{-1} \zeta$, it will cut the locus at $-0.83 \pm j1.89$ (the dominant closed loop poles).



The characteristic equation is $1 + KG(s)H(s) = 0$

$$KG(s)H(s) = -1$$

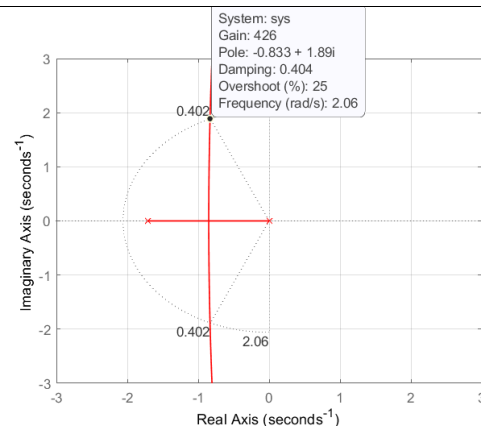
$$K|G(s)H(s)| = 1 \text{ and } \angle G(s)H(s) = \pm(2k+1)180^\circ$$

Since the points $-0.83 \pm j1.89$ are on the rootlocus, they will satisfy the angle condition. From the magnitude condition, let us find out the gain K .

$$K = \frac{1}{|G(s)H(s)|_{s=-0.83 \pm j1.89}} = \frac{1}{\left| \frac{6.63}{s(s+1.71)(s+100)} \right|_{s=-0.83 \pm j1.89}} = \frac{|s(s+1.71)(s+100)|}{6.63} \Big|_{s=-0.83 \pm j1.89}$$

$$K = \left(\frac{|s||s+1.71||s+100|}{6.63} \right)_{s=-0.83 \pm j1.89} = \frac{|-0.83 \pm j1.89| |-0.83 \pm j1.89 + 1.71| |-0.83 \pm j1.89 + 100|}{6.63}$$

$$= \frac{\sqrt{-0.83^2 + 1.89^2} \sqrt{(1.71 - 0.83)^2 + 1.89^2} \sqrt{(100 - 0.83)^2 + 1.89^2}}{6.63} = \frac{\sqrt{4.261} \sqrt{4.3465} \sqrt{9838.26}}{6.63} = 64.38$$



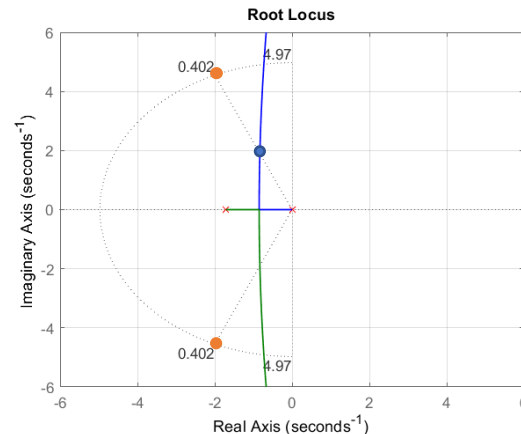
Which is the preamplifier gain we have selected. The closed-loop gain is $6.63 \times 64.38 = 426$ which is visible in the figure.

Suppose we require the following specification for the system $\%OS = 25$, $2\% t_s = 2\text{ s}$, $K_v = 20$. Since the $\%OS$ has not been changed, the value of ζ remains the same.

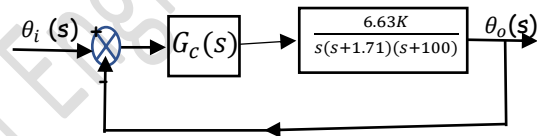
$$t_s = 2 = \frac{4}{\zeta \omega_n} \gggg \zeta \omega_n = 2 \gggg \omega_n = 4.975 \gggg \omega_d = 4.56$$

The new closed-loop pole location is $-\zeta \omega_n \pm j \omega_d = -2 \pm j 4.56$

But, we can see that root locus does not pass through this point. With the given system, we can not achieve the required design specifications. So, we need to modify the parameters of the plant in such a way that the new root locus passes through the desired point. This may not be possible always. Then we need to add additional dynamics in to the system so that the new root locus will pass through the desired points. The additional components that we add for this purpose are known as **compensators**.



Additional poles and zeros are to be introduced in the forward path for the compensation. Consider a system whose transfer function is $G_c(s) = \frac{s+z_c}{s+p_c}$ is added in the forward path. z_c is the compensator zero and p_c is the compensator pole.



Let us assume that $z_c = -2$. Where should we place the compensator pole p_c on the real axis? The new pole location $s = -2 \pm j 4.56$ should satisfy the angle condition considering the open loop poles $p_i = -100, -1.71, 0, p_c$ and the open loop zero $z_c = -2$.

$$\begin{aligned} \theta_1 &= -\tan^{-1} \frac{4.56}{2} = -66.32^\circ = 113.68^\circ \text{ (pole at } s=0) \\ \theta_2 &= -\tan^{-1} \frac{4.56}{2-1.71} = -85.85^\circ = 94.15^\circ \text{ (pole at } s=-1.71) \\ \theta_3 &= \tan^{-1} \frac{4.56}{100-2} = 2.66^\circ \text{ (pole at } s=-100) \\ \theta_4 &=? \text{ (compensator pole)} \end{aligned}$$

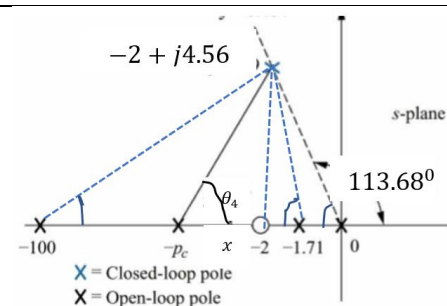
The angle condition is

$$\theta_1 - \theta_2 - \theta_3 - \theta_4 = \pm(2k+1)180$$

$$90 - 113.68 - 94.15 - 2.66 - \theta_4 = -180$$

$$\theta_4 = 59.5^\circ \gggg x = \frac{4.56}{\tan 59.5} = 2.68$$

Hence the compensator pole is at -4.68 .



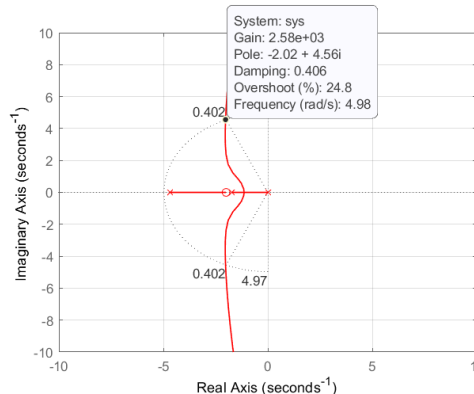
The new forward path transfer function is

$$G(s)H(s) = \frac{6.63K(s+2)}{s(s+1.71)(s+100)(s+4.68)}$$

The closed loop gain is $6.63K = 2580$

$$K = 389$$

We have thus improved the transient response of the system. This is known as **lead compensator**. In lead compensation, the compensator zero is to the right of the compensator pole.



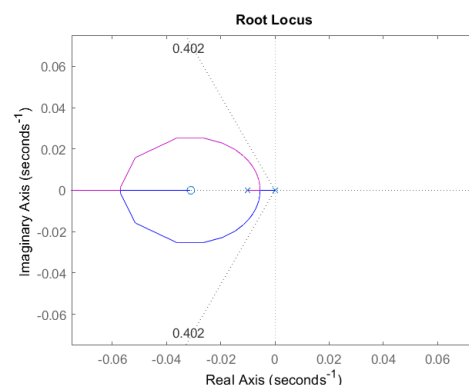
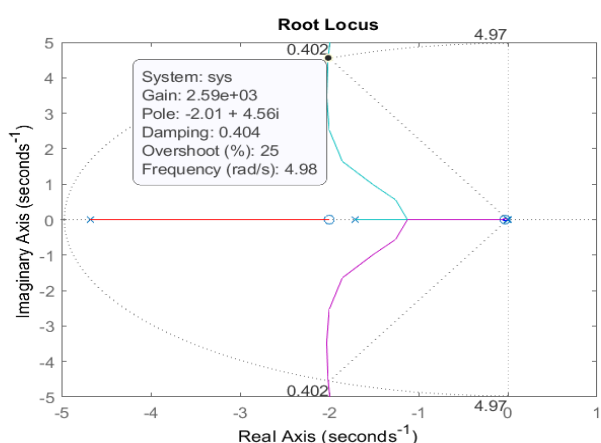
The velocity error constant of the lead compensated system

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \lim_{s \rightarrow 0} s \frac{6.63K(s+2)}{s(s+1.71)(s+100)(s+4.68)} = \frac{2580 \times 2}{1.71 \times 100 \times 4.68} = 6.44$$

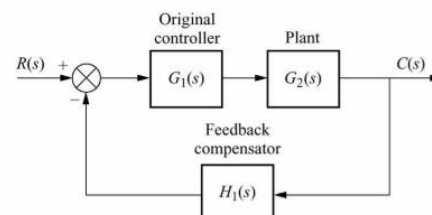
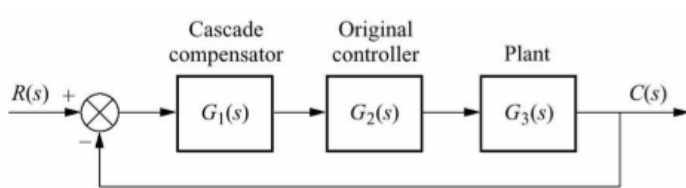
But the required $K_v = 20$. We can notice that the lead compensator did not improve the steady state behavior of the system. To improve the steady state behavior, we add another compensator known as **lag compensator**. Now the entire compensator is known as **lag-lead compensator**. Let us assume a compensator pole close to the origin, $s = -0.01$. We need to improve the steady state behavior by $\frac{20}{6.44} = 3.1$ times. Let us place the zero at $s = -0.031$. Now the transfer function of the lag-lead compensated system becomes

$$G_{LLC}(s)H(s) = \frac{6.63K(s+2)(s+0.031)}{s(s+1.71)(s+100)(s+4.68)(s+0.01)}$$

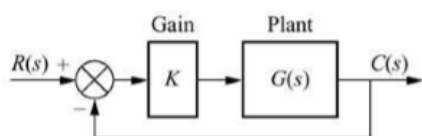
The root loci of the lag-lead compensated system is shown below. The root loci starting from -100 and proceeding to $-\infty$ is not shown. The root loci near the origin is enlarged and shown on the right. The transient behavior is the same but at the same time the steady state behavior is also improved.



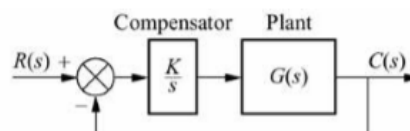
Depending on where the the compensator is placed, they are also known as **cascade compensator** or **feedback compensator**.



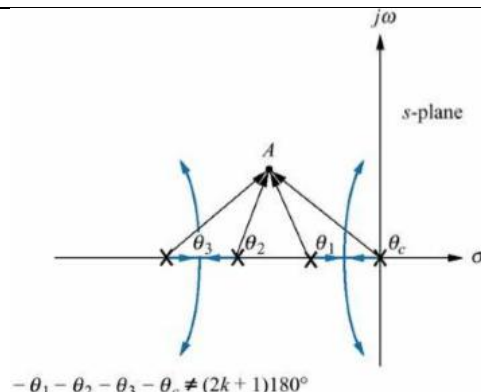
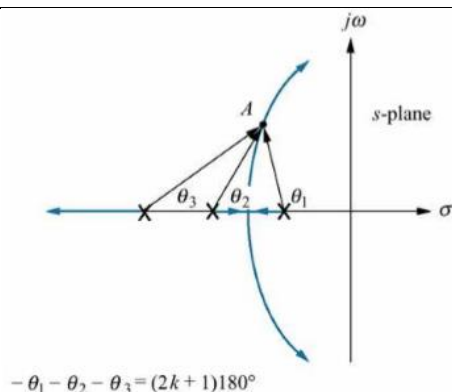
Ideal Integral Controller



Original system



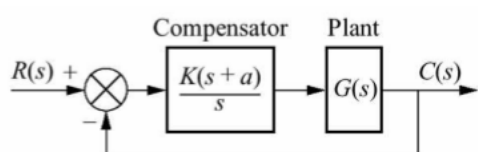
Compensated system



- A pole is added at the origin
- The type of the system is increased. Hence, the steady state error is improved.

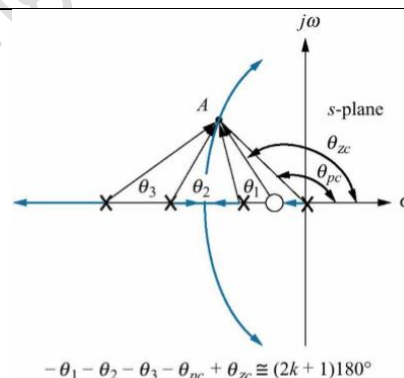
- Desired transient response can not be achieved

Proportional+Integral Controller



$$G_c(s) = K + \frac{K_i}{s}$$

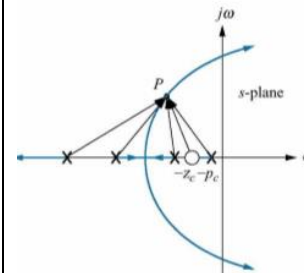
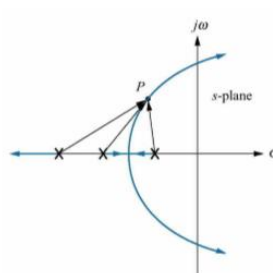
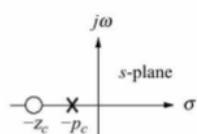
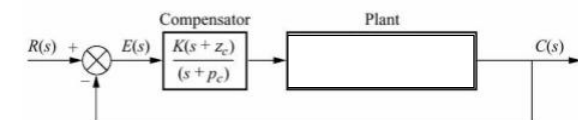
A zero is added because of the proportional part in addition to the pole at the origin



- Steady state error is improved

- Transient response is unaffected.
- Integrator requires active element
- Expensive

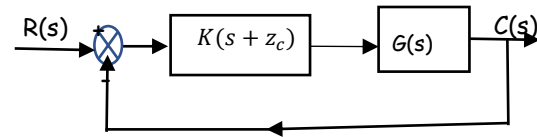
Lag Compensator



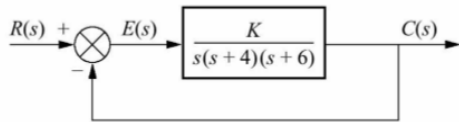
- The pole is not at the origin, but close to the origin to the right of the zero
- Steady state error will be improved. The new error constants will be $\frac{z_c}{p_c}$ times the old ones.
- No effect on the transient response

Proportional+ derivative controller

If the root locus does not pass through the desired points, it needs to be reshaped.

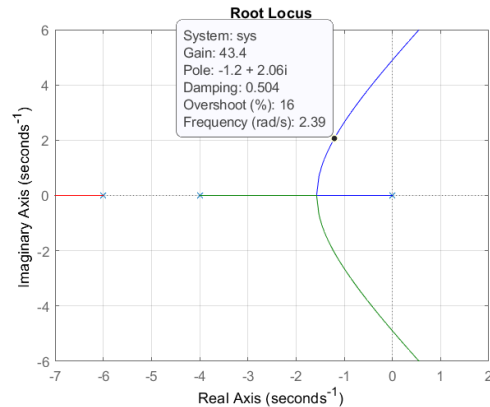


Example.

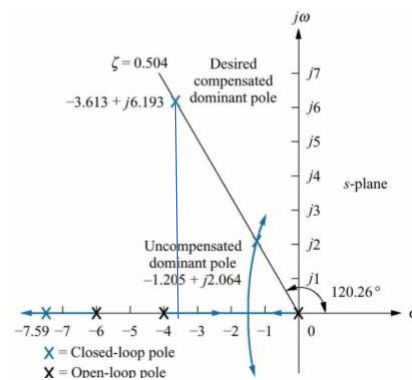


The closed-loop poles for a gain $K = 43.4$ are $-7.59, -1.204 \pm j2.06$. The uncompensated dominant poles are $-1.204 \pm j2.06$. The value of ζ for this dominant poles is 0.504 .

overshoot = 16% , settling time = $\frac{4}{\zeta\omega_n} = 3.32 \text{ s}$



Let us design a PD compensator to yield the same overshoot and three fold reduction in settling time. i.e., $\frac{4}{\zeta\omega_n} = 1.106 \gg \gg \zeta\omega_n = 3.613$. Since the damping factor is the same, we can locate the new pole on the same radial line where the vertical line from $-\zeta\omega_n = -3.613$ intersects.



The angle contribution of poles for the desired pole location

$$\theta_1 = -\tan^{-1} \frac{6.193}{3.613} = -59.74^\circ = 120.26^\circ \text{ (pole at } s = 0 \text{)}$$

$$\theta_2 = \tan^{-1} \frac{6.193}{4-3.613} = 86.42^\circ \text{ (pole at } s = -4 \text{)}$$

$$\theta_3 = \tan^{-1} \frac{6.193}{6-3.613} = 68.9^\circ \text{ (pole at } s = -6 \text{)}$$

$\phi_1 = ??$ (compensator zero at whose location yet to be known)

The angle condition is

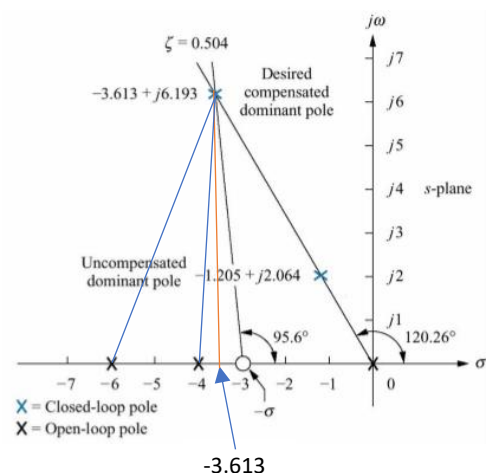
$$\phi_1 - \theta_1 - \theta_2 - \theta_3 = \pm(2k + 1)180$$

$$\phi_1 - 120.26 - 86.42 - 68.9 = -180$$

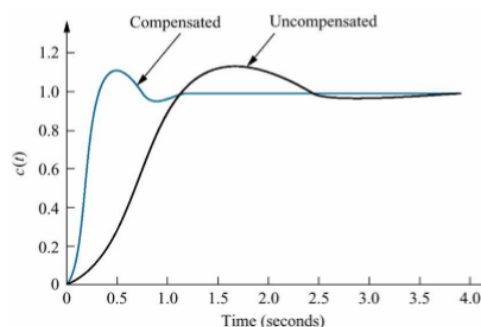
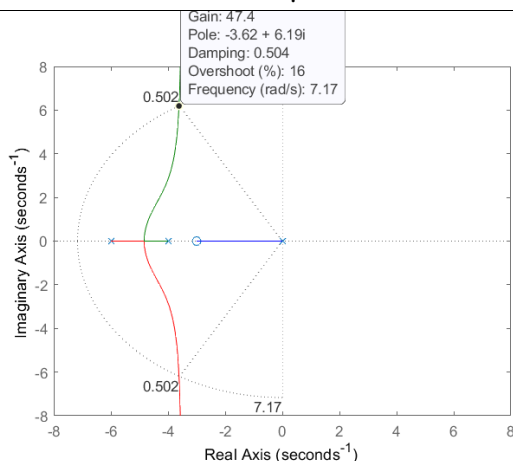
$$\phi_1 = 95.6^\circ = -84.4 \gg \gg x = \frac{6.193}{\tan 84.4} = 0.607$$

Hence the compensator zero should be at

$$-\sigma = -3.613 + 0.607 = -3$$



The root locus after compensation and the transient response are shown below.



- Noise amplification
- Expensive as it requires active elements

Lead Compensator

- Passive element approximation of PD controller, hence cheaper
- Less noise amplification
- Additional pole away from the origin
- $G_c(s) = \frac{s+z_c}{s+p_c}$, compensator pole p_c is away from origin to the left of the compensator zero z_c .
- Choose a compensator zero some where around the open loop poles and using the angle criterion locate the compensator pole. There will be many possibilities and choose the most suitable one considering the static error constants, required gain to reach the design point, and justification of the second order assumption (assumption of dominant poles by looking at the remaining poles).

Proportional+Integral+Derivative (PID) Controller

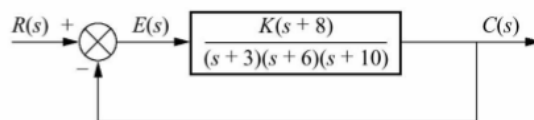
- Improves transient response and steady state behavior
- Uses active elements
- Design Procedure
 - ✓ From the requirements, find out the pole location to meet the transient response specifications.
 - ✓ Design the PD controller
 - ✓ Check the validity of the design.
 - ✓ Design the PI controller to yield the required steady state error value.
 - ✓ Combine PD and PI to obtain K_p , K_i , and K_d

Lag-Lead Compensator

- Improves transient response and steady state behavior
- Uses passive elements
- Design Procedure
 - ✓ find out the pole location to meet the transient response specifications.
 - ✓ Design the lead compensator controller
 - ✓ Evaluate the steady state error and determine the improvement required.
 - ✓ Design the lag compensator

#Exercise 41

Consider the system shown. Design a *PID* controller such that the peak time is $\frac{2}{3}$ times the uncompensated system with 20% overshoot and zero steady state error.

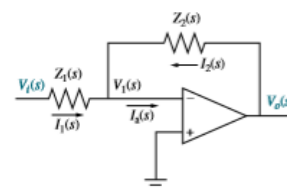


Physical Realisation of compensators

We learnt about op-amps earlier and derived

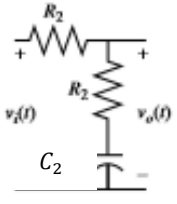
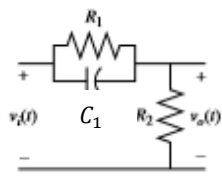
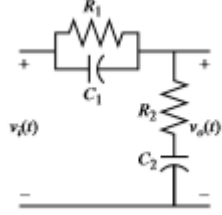
$$\frac{V_0(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

By judiciously choosing $Z_1(s)$ and $Z_2(s)$, this circuit can be used to implement the compensators and controllers

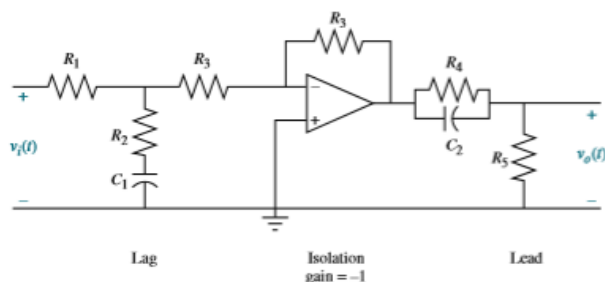


Function	$Z_1(s)$	$Z_2(s)$	$G_c(s) = -\frac{Z_2(s)}{Z_1(s)}$
Gain			$-\frac{R_2}{R_1}$
Integration			$-\frac{1}{RCs}$
Differentiation			$-RCs$
PI controller			$-\frac{R_2}{R_1} \left(s + \frac{1}{R_2 C_2} \right)$
PD controller			$-R_2 C_1 \left(s + \frac{1}{R_1 C_2} \right)$
PID controller			$-\left[\left(\frac{R_2}{R_1} + \frac{C_1}{C_2} \right) + R_2 C_1 s + \frac{1}{R_1 C_2} \right]$
Lag compensator			$-\frac{C_1}{C_2} \left[\frac{s + \frac{1}{R_1 C_1}}{s + \frac{1}{R_2 C_2}} \right]$ where $R_2 C_2 > R_1 C_1$
Lead compensator			$-\frac{C_1}{C_2} \left[\frac{s + \frac{1}{R_1 C_1}}{s + \frac{1}{R_2 C_2}} \right]$ where $R_2 C_2 < R_1 C_1$
Lag-lead compensator can be formed by cascading the lag compensator with the lead compensator as shown			

Lag, lead, and lag-lead compensators can also be implemented with passive network.

Function	Network	Transfer function $\frac{V_o(s)}{V_i(s)}$
Lag compensation		$\left(\frac{R_2}{R_1 + R_2}\right) \frac{\left(s + \frac{1}{R_2 C_2}\right)}{s + \frac{1}{(R_1 + R_2)C_2}}$
Lead compensation		$\frac{\left(s + \frac{1}{R_1 C_1}\right)}{s + \left(\frac{1}{R_1} + \frac{1}{R_2}\right) \frac{1}{C_1}}$
Lag-lead compensation		$\frac{\left(s + \frac{1}{R_1 C_1}\right) \left(s + \frac{1}{R_2 C_2}\right)}{s^2 + s \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1}\right) + \frac{1}{R_1 C_1 R_2 C_2}}$

A lag-lead compensator can be realised by cascading the lead network with the lag network. When we use this, care should be taken to isolate one from the other. Otherwise, one network will load the other and the transfer function will not be the product of the individual transfer functions. A possible way of achieving this by using an op-amp for isolation.



References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons